

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - HIGH (Pseudo-code Writing)

COURSE CODE: CSA09 COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO1: *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3, BL4)*

Assessment Tool Weightage: 20%

Total Marks: 50

QUESTIONS: 5*Code of Conduct:*

ANNEXURE A

PSEUDO-CODE WRITING

I S.K.Deepa Arathi (192571020) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
LIST OF QUESTIOONS

1. WRITE A JAVA PROGRAM OF SUM AND AVERAGE OF FIVE NUMBERS AND GET THE INPUT FROM THE USERS.

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QUESTIONS

Q1
Sum and Average
10 marks

Q2
Days Conversion
10 marks

Q3
Voting Eligibility
10 marks

Q4
Income Tax
10 marks

Q5
Armstrong Number
10 marks

5/5 answered

Q1 Sum and Average 10 marks

Write a Java program to find the sum and average of five numbers. Get the input from the user.

Your Code

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6         int sum = 0;  
7         int[] nums = new int[5];  
8         for (int i = 0; i < 5; i++) {  
9             nums[i] = sc.nextInt();  
10            sum += nums[i];  
11        }  
12        double average = sum / 5.0;  
13        System.out.println("Sum = " + sum);  
14        System.out.println("Average = " + average);  
15    }  
16 }
```

Input

1
4
56
78
--

Output

Sum = 228
Average = 45.6

Visible Test Cases

Test Case 1
Test Case 2
Test Case 3

0 / 10 marks

2.WRIT A JAVA PROGRAM FOR CONVERT NUMBER OF DAYS IN TERM OF
YEA



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QUESTIONS

Q1

Sum and Average

10 marks

✓

Q2

Days Conversion

10 marks

✓

Q3

Voting Eligibility

10 marks

✓

Q4

Income Tax

10 marks

✓

Q5

Armstrong Number

10 marks

✓

5/5 answered

Q2 Days Conversion

10 marks

Write a Java program to convert the given number of days into years, months, weeks and days.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int totalDays = sc.nextInt();
7
8         int years = totalDays / 365;
9         totalDays = totalDays % 365;
10
11         int months = totalDays / 30;
12         totalDays = totalDays % 30;
13
14         int weeks = totalDays / 7;
15         int days = totalDays % 7;
16
17         System.out.println("Years: " + years);
18         System.out.println("Months: " + months);
19         System.out.println("Weeks: " + weeks);
20         System.out.println("Days: " + days);
21     }
22 }
```

Input

4578

Output

Years: 12
Months: 6
Weeks: 2
Days: 4

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

0 / 10 marks

RS, MONTHS, WEEK AND DAYS.

3. WRITE A JAVA PROGRAM FOR FIND THE GIVEN PERSON IS ELIGIBLE FOR VOTING OR NOT, IF NOT ELEGIBLE HOW MANY YEARS TO BE LEFT.

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QUESTIONS

Q1
Sum and Average
10 marks

Q2
Days Conversion
10 marks

Q3
Voting Eligibility
10 marks

Q4
Income Tax
10 marks

Q5
Armstrong Number
10 marks

5/5 answered

Q3 Voting Eligibility 10 marks

Write a Java program to check whether a person is eligible for voting. If not, display how many years are left.

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         if (!sc.hasNextInt()) return;
7         int age = sc.nextInt();
8
9         if (age >= 18) {
10             System.out.print("Eligible for voting");
11         } else {
12             int yearsLeft = 18 - age;
13             System.out.print("Not eligible. " + yearsLeft + " years left");
14         }
15     }
16 }
```

Input

18

Output

Eligible for voting

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

0 / 10 marks

WRITE A PSUEDO CODE FOR PREPARATION OF INCOME TAX FOR THE FOLLOWING CONDITIONS

ANNUAL INCOME TAX(RS)

<500000 0

5L TO 7.5 L 5%

7.5L TO 10L 7.5%

10L TO 15L 10%

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Armstrong Number
10 marks
5/5 answered

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6  
7         int empNo = sc.nextInt();  
8         String empName = sc.next();  
9         double salary = sc.nextDouble();  
10  
11         double tax;  
12         if (salary < 500000) {  
13             tax = 0;  
14         } else if (salary <= 750000) {  
15             tax = 0.05 * salary;  
16         } else if (salary <= 1000000) {  
17             tax = 0.075 * salary;  
18         } else if (salary <= 1500000) {  
19             tax = 0.10 * salary;  
20         } else if (salary <= 2000000) {  
21             tax = 0.15 * salary;  
22         } else {  
23             tax = 0.20 * salary;  
24         }  
25  
26         System.out.printf("Employee No: %d\n", empNo);  
27         System.out.println("Employee Name: " + empName);  
28         System.out.printf("Salary: %.2f\n", salary);  
29         System.out.printf("Income Tax: %.2f\n", tax);  
30     }  
31 }
```

101
DEEPA
50000

Output
Employee No: 101
Employee Name: DEEPA
Salary: 50000.00
Income Tax: 0.00

Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
0 / 10 marks

5. WRITE A PSUEDO CODE FIND THE GIVEN NUMBER IS ARMSTRONG NUMBER OR NOT.



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Days Conversion
10 marks

Q3
Voting Eligibility
10 marks

Q4
Income Tax
10 marks

Q5
Armstrong Number
10 marks

5/5 answered

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         if (!sc.hasNextLong()) return;
7
8         long n = sc.nextLong();
9
10        if (n < 0) {
11            System.out.print("Not Armstrong");
12            return;
13        }
14
15        int digits = 0;
16        long temp = n;
17
18        if (temp == 0) digits = 1;
19        while (temp > 0) {
20            digits++;
21            temp /= 10;
22        }
23
24        temp = n;
25        long sum = 0;
26
27        while (temp > 0) {
28            int digit = (int) (temp % 10);
29            sum += powInt(digit, digits);
30        }
```

151

Output
Not Armstrong

Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
0 / 10 marks

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient